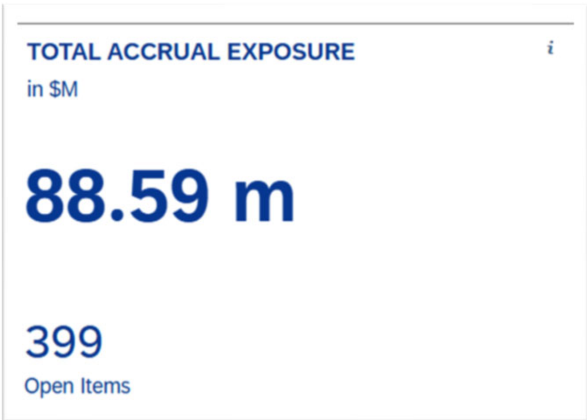


Total Accrual Exposure

Definition: This is based on real time values - total monetary value of Goods Receipts that have not yet been fully matched by Supplier Invoices as of the current date — representing the outstanding financial liability that needs to be accrued on the balance sheet.

PREDICTED PERIOD END ACCRUAL

in $M

61.88 m

Databricks ML Perspective

ML Formula

Predicted Contribution = Exposure Amount × Predicted Probability

Overall Forecast = Σ(Predicted Contribution)

Features Used

- GR Age
- Historical Invoice Behavior
- Invoice Coverage
- Signed Amount
- Fiscal Timing
- Supplier History

ML Flow

SAP Historical Data → Feature Engineering → Random Forest Model → Probability → Forecast

SAP Functional Perspective

Business Formula

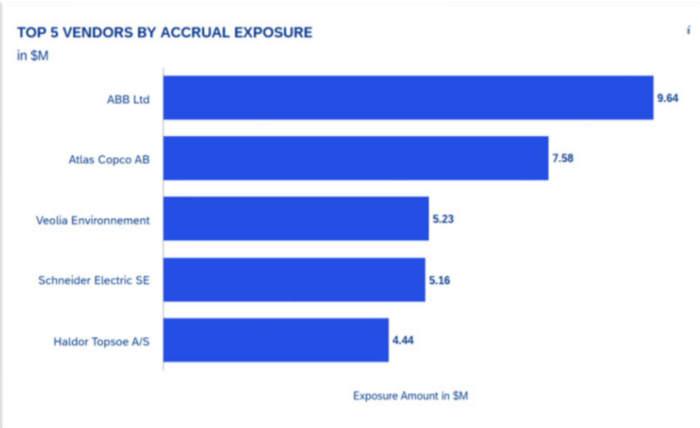
Predicted Period End Accrual = Σ(Open Accrual Exposure × Probability of Remaining Open)

Business Variables

- Open GR/IR Amount – Current outstanding accrual
- GR Aging – Older items are less likely to close
- Historical Invoice Timeliness – Vendor behavior
- Invoice Coverage % – Portion already invoiced
- Signed GR/IR Amount – Net financial exposure

Simple Explanation

1. Read all open GR/IR items.
2. Evaluate business factors.
3. Estimate which items remain open.
4. Sum predicted open balances.



Top 5 Vendors by Accrual Exposure

Step 1: For each open transaction, calculate Exposure Amount (USD) = (GR Amount − IR Amount) × Exchange Rate.

Step 2: Group all open transactions by Supplier (Vendor).

Step 3: For each supplier, calculate total Vendor Exposure = sum of all exposure amounts for that supplier.

Step 4: Sort all suppliers in descending order based on total exposure.

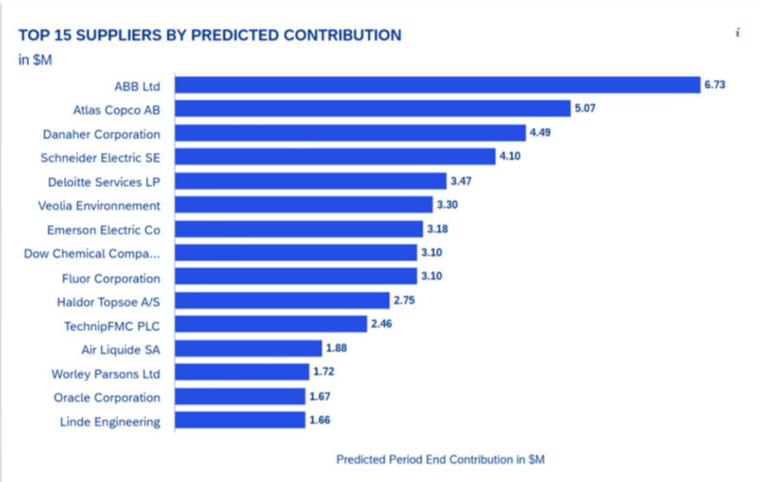
Step 5: Select the top 5 suppliers with the highest exposure.

Supporting Metrics

Open Items per Vendor = Count of transactions.

Average Days Open = Average of days open for transactions.

% Contribution = (Vendor Exposure / Total Exposure of all vendors) × 100.



SAP Functional Perspective

Business Formula
Supplier Contribution = Σ(Predicted Contribution by Supplier)

Business Variables
- Supplier
- Open GR/IR Amount
- Probability of Remaining Open
- Predicted Contribution

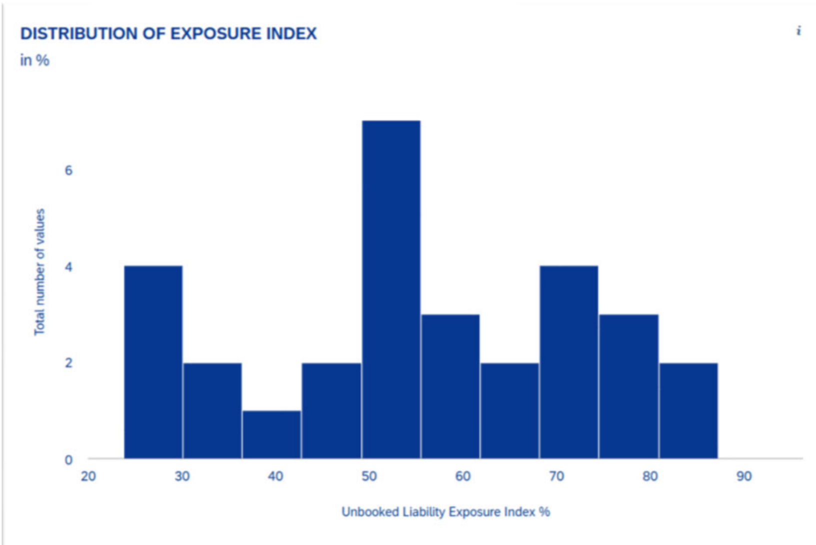
Business Logic
1. Group open accruals by supplier.
2. Sum predicted contribution.
3. Rank suppliers highest to lowest.
4. Display Top 15 suppliers

Databricks ML Perspective

ML Formula
Predicted Contribution = Exposure × Probability
Supplier Contribution = Σ(Predicted Contribution)

ML Features
- Exposure Amount
- Supplier History
- Invoice Behavior
- Aging
- Fiscal Timing

ML Flow
Predict each accrual → Aggregate by supplier → Sort descending → Show Top 15



A frequency distribution showing how many suppliers fall into each exposure index percentage band, revealing whether Unbooked liability risk is concentrated in a few vendors or spread systemically across the portfolio.

SAP Functional Perspective
Business Formula
Exposure Index (%) = (Estimated Unbooked Liability ÷ Total Spend) × 100

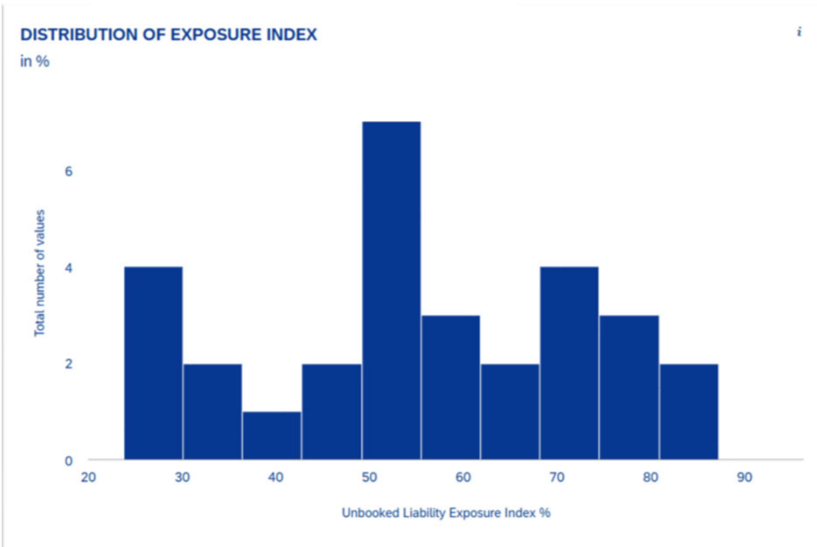
Business Variables
• Open GR/IR Amount
• Predicted No-Document Accrual
• Total Organizational Spend
• Supplier

Business Logic
1. Calculate expected Unbooked liability.
2. Divide by total spend.
3. Convert to percentage.
4. Display supplier distribution.

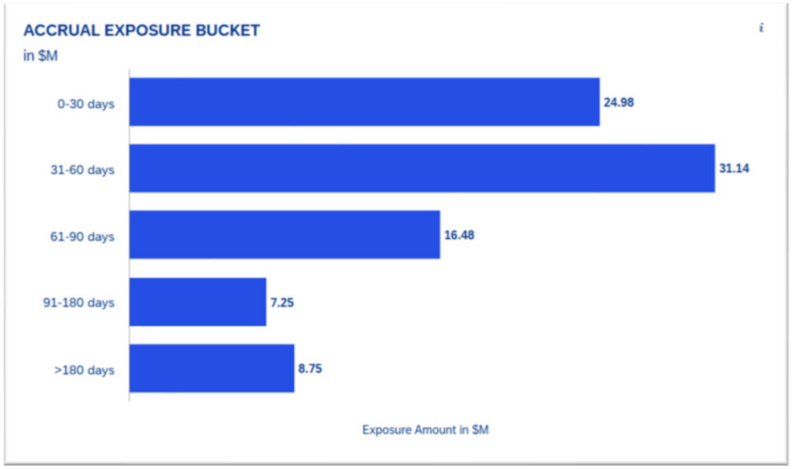
Databricks ML Perspective
ML Formula
Estimated Liability = GR/IR + Predicted No-Document Accrual
Exposure Index = Liability ÷ Spend × 100

ML Features
• PO History
• GR/IR Exposure
• Invoice Behavior
• Supplier Pattern
• Fiscal Timing

ML Flow
Predict hidden liabilities → Calculate Exposure Index → Group suppliers into ranges → Display histogram

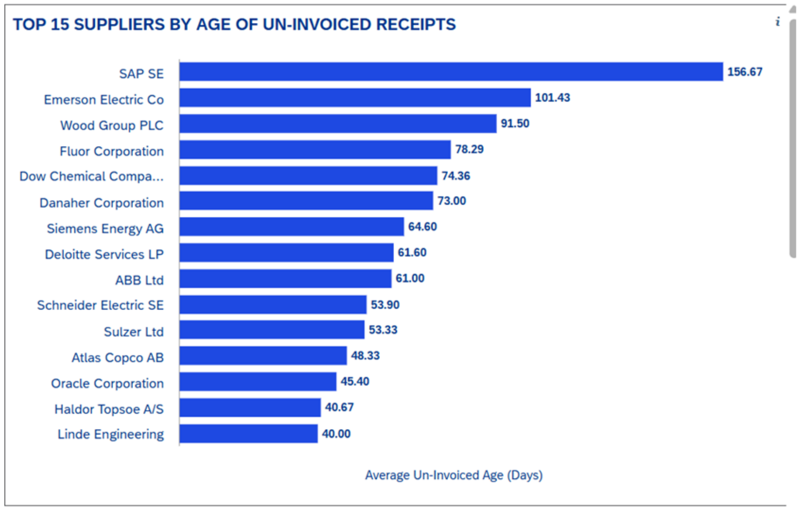


Unbooked Liability Exposure Index
Standard accounting only books a liability once a Goods Receipt (GR) is posted — i.e., once the system knows goods arrived. But what about situations where goods or services were consumed but nobody posted the GR? Those liabilities are completely invisible to normal finance reporting. The model predicts the dollar value of those *hidden* liabilities for every supplier. It then adds them to the known GR/IR accruals and asks: *what percentage of everything we owe this supplier is sitting unresolved?* That percentage is the **Exposure Index**. A supplier at 85% means nearly all your financial relationship with them is in limbo — a red flag for the CFO and the procurement team. Suppliers are then bucketed as High (>80%), Medium (50–80%), or Low (<50%) risk for action prioritization.



Accrual Exposure Bucketing (Aging)
Step 1: Get input data for each open item, take: Days_Open
Step 2: Assign aging bucket For each item:
If Days_Open ≤ 30 → assign "0–30 days
Else if Days_Open ≤ 60 → assign "31–60 days
Else if Days_Open ≤ 90 → assign "61–90 days
Else if Days_Open ≤ 180 → assign "91–180 days
Else → assign ">180 days
Step 3: Check overdue status For each item : If Days_Open > 30 → mark Overdue = "Yes“ Else → mark Overdue = "No“
Step 4: Calculate metrics per bucketFor each bucket : Item Count = number of items
Total Exposure = sum of amounts
% of Total Exposure = (Bucket Exposure / Total Exposure) × 100
Step 5: Calculate overall metric Beyond Threshold Rate =(Number of items with Days_Open > 30 / Total items) × 100

Group items by how old they are, mark overdue items, and calculate count and exposure for each age group.



CFO Business Value

- Identifies suppliers with persistent invoicing delays.
- Highlights ageing accrual risks.
- Helps Accounts Payable prioritize vendor follow-up.
- Reduces month-end accrual uncertainty and improves financial accuracy.

Top 15 Suppliers by Average Un-Invoiced Receipt Age

SAP Functional Perspective

Business Purpose

Identify suppliers whose Goods Receipts (GR) remain un-invoiced for the longest period, helping Finance focus on vendors causing aged accruals.

Business Formula

Average Un-Invoiced Age (Days) = Σ(Days Open for Un-Invoiced Receipts) / Number of Un-Invoiced Receipts

Where:

Days Open = Current Date – Goods Receipt Date

SAP Business Variables

Variable	Description
Supplier	Vendor being analyzed
Goods Receipt Date	Date when goods were received
Current / Reference Date	Date of report execution
Days Open	Age of the un-invoiced receipt
Number of Un-Invoiced Receipts	Total open GRs without invoices

Business Logic

- Select Goods Receipts with no supplier invoice.
- Calculate the age of each receipt (Current Date – Goods Receipt Date).
- Group records by supplier.
- Calculate the average un-invoiced age for each supplier.
- Rank suppliers from highest average age to lowest.
- Display the Top 15 suppliers.

Business Logic

- Select all open GR/IR items.
- Group records by Goods Receipt posting month.
- Sum the exposure amount for each month.
- Count the number of open PO items for each month.
- Display the monthly trend for the last 12 months.

CFO Business Value

- Highlights whether accrual exposure is increasing or decreasing over time.
- Identifies seasonal peaks and recurring business patterns.
- Supports period-end forecasting and accrual planning.
- Helps Finance monitor the effectiveness of invoice processing and accrual clearance.

Total Accrual Exposure in $ (12-Month Rolling Trend):Real-time measure of the total open GR/IR accrual liability amount tracked continuously over a rolling 12-month period to monitor exposure trends and operational backlog.

Total Accrual Exposure – 12 Month Rolling Trend

SAP Functional Perspective

Business Purpose

Track the monthly trend of total open GR/IR accrual exposure over the last 12 months. This helps Finance identify seasonal patterns, increasing liabilities and the effectiveness of accrual clearing.

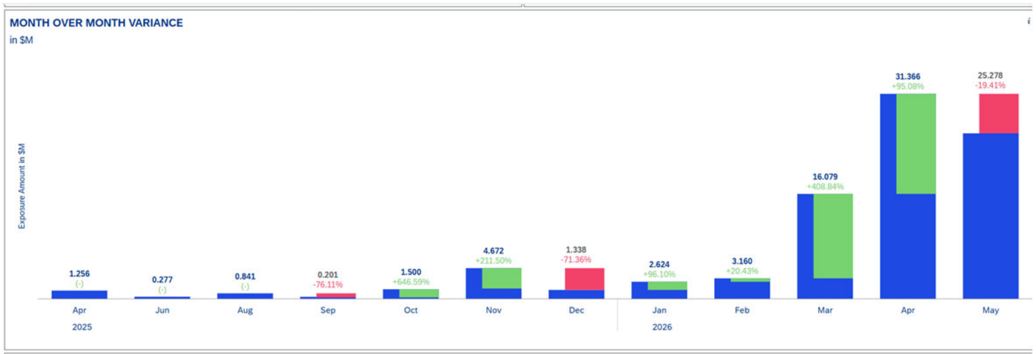
Business Formula

Monthly Accrual Exposure = SUM(Open GR/IR Exposure Amount)

Grouped By: Goods Receipt Posting Month

SAP Business Variables

Variable	Description
Goods Receipt Date	Month in which the GR was posted
Exposure Amount	Open GR/IR liability amount
Posting Month	Month used for trend analysis
Open PO Count	Number of open purchase order items
Reference Period	Rolling 12-month reporting period



Variable	Description
Current Month Exposure	Total open GR/IR exposure for the current month
Previous Month Exposure	Total open GR/IR exposure for the previous month
Exposure Amount	Outstanding GR/IR liability
Posting Month	Month used for comparison
MoM Variance %	Percentage increase or decrease compared to the previous month

Month-on-Month (MoM) Variance %

SAP Functional Perspective

Business Purpose

Measure the percentage change in total open accrual exposure between consecutive months. This helps Finance quickly identify unusual increases or decreases that require investigation.

Business Formula

MoM Variance (%) = ((Current Month Exposure - Previous Month Exposure) / Previous Month Exposure) × 100

Where:

Monthly Exposure = SUM(Open GR/IR Exposure Amount)

Business Logic

• Calculate the total open accrual exposure for each month.
• Compare the current month's exposure with the previous month.
• Calculate the percentage variance.
• Highlight significant increases or decreases.
• Display the monthly trend for Finance review.

CFO Business Value

• Detects unusual spikes or drops in accrual exposure.
• Highlights potential procurement or invoice processing issues.
• Supports proactive month-end planning.
• Helps Finance investigate abnormal movements before financial close.

Average Days Open by Worklist Category
SAP Functional Perspective
Business Purpose
Monitor the average number of days that open Goods Receipt (GR) items remain un-invoiced each month. This KPI helps Finance measure how quickly invoices are being received and processed.
Business Formula
Average Days Open = SUM(Days Open for Un-Invoiced GR Items) / Number of Un-Invoiced GR Items

Where:
Days Open = Reference Date - Goods Receipt Date

Business Logic
- Select all Goods Receipts with no supplier invoice.
- Calculate the age of each open GR item.
- Group records by month and worklist category.
- Calculate the average days open for each month.
- Display the monthly trend to monitor process performance.

CFO Business Value
- Measures the efficiency of invoice processing.
- Identifies trends in ageing open GR items.
- Highlights delays requiring operational attention.
- Supports continuous improvement of the procure-to-pay process.

Real-time measure of the average number of days open GR/IR worklist items (by category such as invoice missing, price variance, and quantity variance) remain unresolved, tracked continuously over a 12-month rolling period to monitor processing efficiency and backlog aging.

Variable	Description
Goods Receipt Date	Date on which goods were received
Reference Date	Date used to calculate ageing
Days Open	Number of days the GR remains un-invoiced
Worklist Category	Business category used for reporting
Un-Invoiced GR Count	Total open GR items without invoices

AUTO - POST ELIGIBLE	EXCEPTION REVIEW QUEUE	WATCHLIST / OBSERVATION QUEUE	NO ACTION	TOTAL ACCRUAL EXPOSURE in $
235 Open Items	153 Open Items	5 Open Items	14 Open Items	60.49 m 407 Open Items

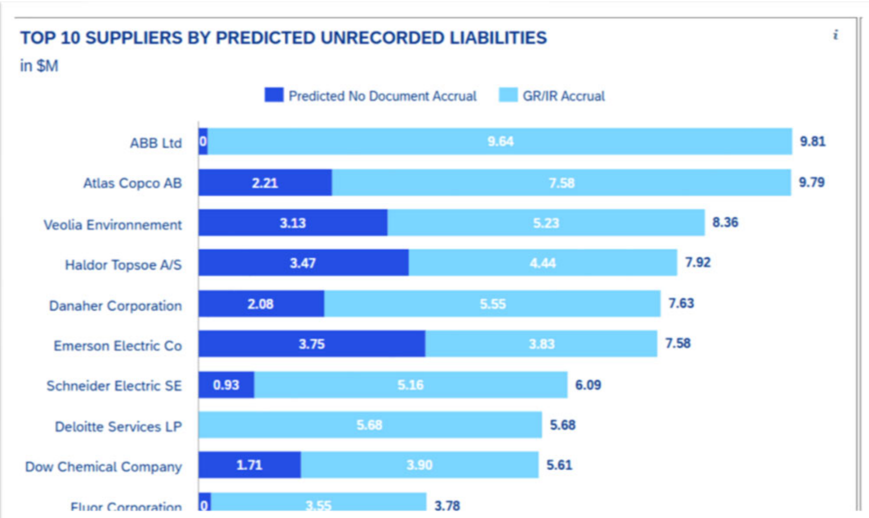
RECOMMENDED ACCRUAL ITEMS

Measures

Purchase Order	Purchase Order Item	Purchase Order Item Text	Supplier	G/L Account	Cost Center	Profit Center	Fiscal_Year	Fiscal Period	Worklist Category	RCA	Recommended Action
4500100004	10	Consulting - Digital Twin	0050100012	65000000	10106482	YB300	2024	11	Cross-Period GR-IR	Cross-Period Cut-Off: GR posted in period 2024/11 but IR posted in later period 2025/01. Exposure $165952.56 USD requires period-end accrual adjustment.	POST
4500100004	20	Water Treatment Chemicals	0050100012	65000000	10102298	YB120	2024	11	Cross-Period GR-IR	Cross-Period Cut-Off: GR posted in period 2024/11 but IR posted in later period 2024/12. Exposure $861243.06 USD requires period-end accrual adjustment.	POST
4500100006	20	Rotating Equipment Repair	0050100020	64000000	10101179	YB120	2024	12	Cross-Period GR-IR	Cross-Period Cut-Off: GR posted in period 2024/12 but IR posted in later period 2025/01. Exposure $110215.4 USD	POST

Measures

Purchase Order	Purchase Order Item	Purchase Order Item Text	Supplier	G/L Account	Cost Center	Profit Center	Fiscal_Year	Fiscal Period	Worklist Category	RCA	Recommended Action
4500100011	10	IT Infrastructure Upgrade	0050100008	64000000	10104513	YB300	2026	1	Cross-Period GR-IR	Cross-Period Cut-Off: GR posted in period 2026/01 but IR posted in later period 2026/03. Exposure $206793.3 USD requires period-end accrual adjustment.	POST
4500100015	20	Nitrogen Supply Contract	0050100016	51100000	10107413	YB200	2024	9	Invoice Missing	Invoice Missing: Goods receipt posted (GR amount $137354.1 USD) but no corresponding invoice receipt (RE) exists. Open for 616 days. Likely cause: vendor invoice delayed, lost, or not yet submitted.	POST
4500100028	10	Boiler Tube Inspection	0050100005	62000000	10105112	YB110	2025	10	Quantity Variance ...	Quantity Variance: GR qty (1000.0) exceeds IR qty (383.0). Unmatched qty: 617.0 units @ 232.27 = exposure $154775.44 USD. Likely cause: partial delivery invoiced or GR over-receipt.	POST
4500100031	10	Rotating Equipment Repair	0050100007	64000000	10100650	YB300	2024	5	Invoice Missing	Invoice Missing: Goods receipt posted (GR amount $623789.29 USD) but no corresponding invoice receipt (RE) exists. Open for 746 days. Likely cause: vendor invoice delayed, lost, or not yet submitted.	POST

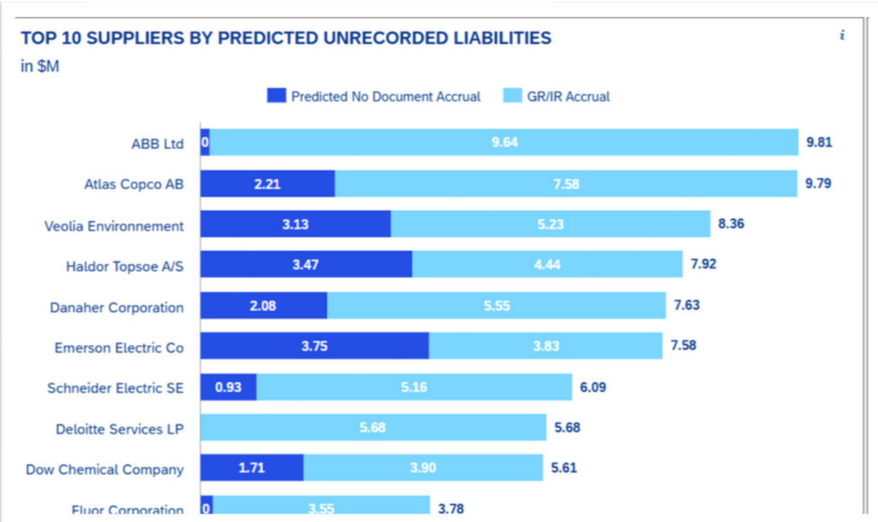


Ranks the 10 suppliers with the highest total predicted unrecorded liabilities, split into the known rule-based GR/IR accrual and the ML-predicted no-document accrual — showing both visible and hidden liability components per vendor.

Variable	Description
Vendor	Vendor being analysed
Open GR/IR Exposure	Current outstanding liability
Goods Receipt Amount	Value of goods received
Invoice Status	Whether supplier invoice has been received
Total Liability	Combined actual and predicted liability

SAP Functional Perspective (Business Logic)
Business Purpose
Identify the suppliers expected to have the highest total liability exposure by combining current open GR/IR balances with predicted unrecorded liabilities.

Business Formula
Total Supplier Liability = Open GR/IR Exposure + Predicted Unrecorded Liability
Business Logic
- Retrieve all open GR/IR balances.
- Group data by supplier.
- Add the predicted unrecorded liabilities.
- Calculate the total liability for each supplier.
- Rank suppliers from highest to lowest.
- Display the Top 10 suppliers.
- Light GBM prediction model



ML Formula

Predicted Liability = Predicted No-Document Accrual
Total Supplier Liability = Open GR/IR Exposure + Predicted Liability

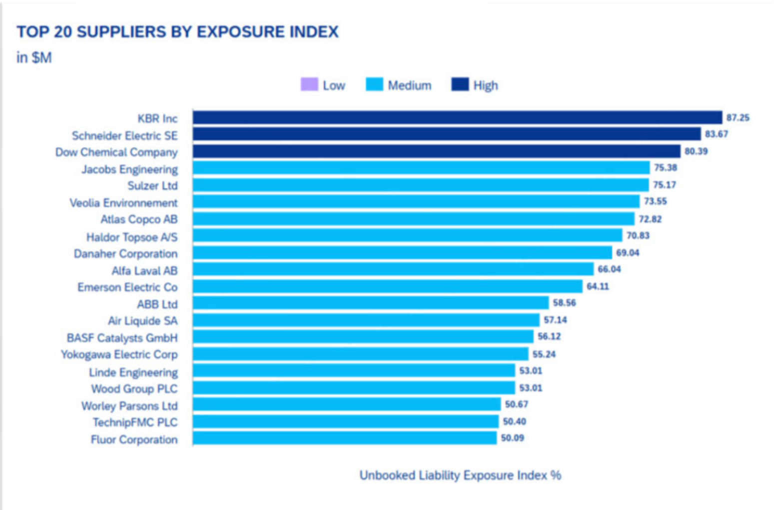
ML Process

- Load historical SAP data.
- Perform feature engineering.
- Run the Light GBM prediction model.
- Predict unrecorded liabilities.
- Combine predictions with open GR/IR exposure.
- Rank suppliers and display the Top 10 dashboard.

Executive Message

SAP provides the current open GR/IR exposure for each supplier. Databricks predicts additional liabilities where invoices are expected but not yet received. Combining both values gives Finance a complete supplier liability view and helps prioritize vendor follow-up before period close.

ML Feature	Business Meaning
Goods Receipt Amount	Value of goods received
PO Attributes	Purchase order characteristics
Supplier History	Historical invoicing behavior
Goods Receipt Age	Time since goods receipt
Invoice Pattern	Previous invoice timing
Fiscal Period	Month-end behavior



SAP Business Variable	Description
Supplier	Vendor being analysed
Open GR/IR Exposure	Current outstanding liability
Predicted Unrecorded Liability	Predicted liability from ML model
Total Organisational Spend	Supplier spend used for normalisation
Exposure Index %	Liability as a percentage of supplier spend

Top 20 Suppliers by Exposure Index %

SAP Functional Perspective (Business Logic)

Business Purpose

Rank suppliers based on their Exposure Index (%) to identify vendors contributing the highest proportion of unresolved liability relative to their organizational spend-values a calculated in real time .

Business Formula

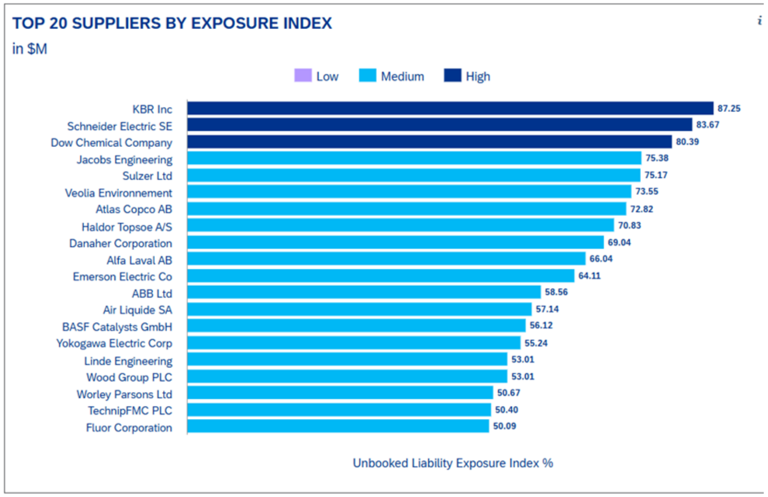
Exposure Index (%) = (Total Supplier Liability / Total Organizational Spend) × 100

Where:

Total Supplier Liability = Open GR/IR Exposure + Predicted Unrecorded Liability

Business Logic

- Retrieve supplier exposure from SAP.
- Add predicted unrecorded liabilities.
- Calculate total supplier liability.
- Calculate the Exposure Index percentage.
- Rank suppliers by Exposure Index in descending order.
- Display the Top 20 suppliers.



ML Feature	Business Meaning
Goods Receipt Amount	Value of goods received
PO Attributes	Purchase order characteristics
Supplier History	Historical invoicing behavior
Goods Receipt Age	Age of open receipts
Invoice Pattern	Historical invoice timing
Total Organizational Spend	Used to calculate Exposure Index

Databricks ML Perspective (Prediction Logic)

ML Formula

Predicted Liability = Predicted No-Document Accrual

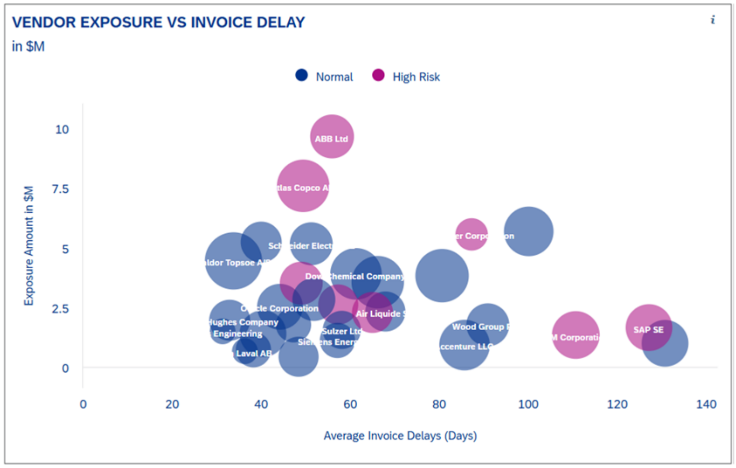
Exposure Index (%) = ((Open GR/IR Exposure + Predicted Liability) / Total Organizational Spend) × 100

ML Process

- Load historical SAP data.
- Generate ML features.
- Predict unrecorded liabilities using the ML model.
- Combine predictions with open GR/IR exposure.
- Calculate Exposure Index.
- Rank suppliers and display the Top 20 dashboard.

Executive Message

SAP provides the current supplier exposure, while Databricks predicts additional unrecorded liabilities. The Exposure Index normalizes total liability against supplier spend, enabling Finance to identify suppliers with disproportionately high financial risk and priorities corrective actions.



SAP Business Variable	Description
Supplier	Vendor being analysed
Goods Receipt Date	Date goods were received
Reference Date	Date used for ageing calculation
Invoice Delay (Days)	Days since GR without invoice
Open GR/IR Exposure	Outstanding liability amount
Open GR Count	Number of open GR items

Vendor Exposure vs Invoice Delay

SAP Functional Perspective (Business Logic)

Business Purpose

Compare supplier invoice delays with the corresponding open liability exposure to identify vendors that have both high financial impact and long invoice processing times.

Business Formula

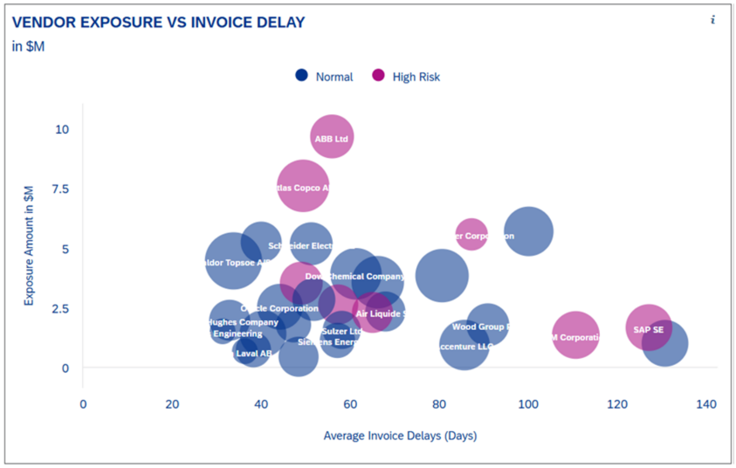
Invoice Delay (Days) = Reference Date - Goods Receipt Date

Exposure Amount = SUM(Open GR/IR Exposure)

Bubble Size = Number of Open GR Items

Business Logic

- Select all open GR/IR items with no supplier invoice.
- Calculate invoice delay for each item.
- Sum exposure amount by supplier.
- Count open GR items for each supplier.
- Plot suppliers on a bubble chart using Delay (X-axis), Exposure (Y-axis) and Open GR Count (bubble size).



Business Meaning	Description
Invoice Delay	Age of un-invoiced receipts
Exposure Amount	Financial impact
Supplier History	Historical invoicing behavior
Open GR Count	Volume of pending items
Goods Receipt Age	Time since receipt
Fiscal Period	Month-end processing pattern

Databricks ML Perspective (Prediction Logic)

ML Formula

Risk Score = f(Invoice Delay, Exposure Amount, Supplier Behavior)

Bubble Size = Open GR Count

Model Features Used

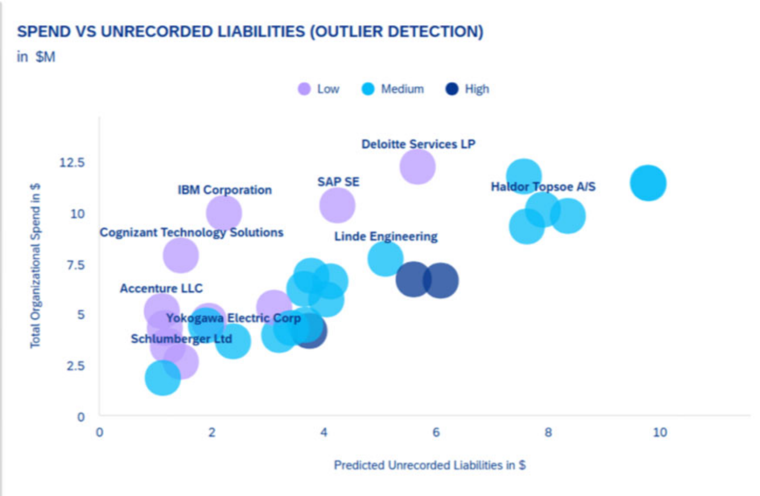
ML Feature

ML Process

- Load historical SAP data.
- Prepare supplier features.
- Evaluate supplier risk using the ML model.
- Calculate exposure and delay metrics.
- Display suppliers on the bubble chart to prioritise high-risk vendors.

Executive Message

SAP provides supplier exposure and invoice ageing information, while Databricks analyses historical supplier behaviour to highlight vendors with the greatest combination of financial exposure and invoice delay. This enables Finance to prioritise supplier follow-up before period close.



SAP Business Variable	Description
Supplier	Vendor being analysed
Open GR/IR Exposure	Current outstanding liability
Total Organizational Spend	Supplier spend
Open GR Count	Number of open GR items
Total Unrecorded Liability	Combined actual and predicted liability

Spend vs Unrecorded Liabilities (Outlier Detection)

SAP Functional Perspective (Business Logic)

Business Purpose

Compare supplier spend with total unrecorded liabilities to identify suppliers whose liability exposure is unusually high relative to their spend and highlight potential outliers for Finance review.

Business Formula

Total Unrecorded Liability = Open GR/IR Exposure + Predicted Unrecorded Liability

Business Logic

- Retrieve supplier spend and open GR/IR balances.
- Add predicted unrecorded liabilities.
- Calculate total supplier liability.
- Plot Liability (X-axis), Spend (Y-axis) and Open GR Count (bubble size).
- Highlight suppliers with unusually high liability relative to spend.

Databricks ML Perspective (Prediction Logic)

ML Formula

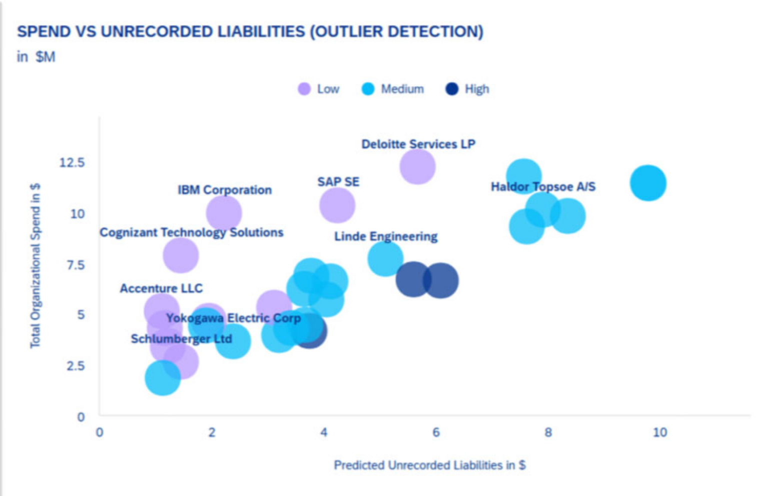
Predicted Liability = Predicted No-Document Accrual

Total org spend = Po spend total

Total Unrecorded Liability = Open GR/IR Exposure + Predicted Liability

Unbooked Liability Exposure Index

Standard accounting only books a liability once a Goods Receipt (GR) is posted — i.e., once the system knows goods arrived. But what about situations where goods or services were consumed but nobody posted the GR? Those liabilities are completely invisible to normal finance reporting. The model predicts the dollar value of those *hidden* liabilities for every supplier. It then adds them to the known GR/IR accruals and asks: *what percentage of everything we owe this supplier is sitting unresolved?* That percentage is the **Exposure Index**. A supplier at 85% means nearly all your financial relationship with them is in limbo — a red flag for the CFO and the procurement team. Suppliers are then bucketed as High (>80%), Medium (50–80%), or Low (<50%) risk for action prioritization.



ML Feature	Business Meaning
Goods Receipt Amount	Value of goods received
Supplier Spend	Supplier purchasing volume
Supplier History	Historical invoicing behaviour
Goods Receipt Age	Age of pending receipts
Invoice Pattern	Past invoice timing
Open GR Count	Volume of unresolved items

ML Process

- Load historical SAP data.
- Create supplier-level features.
- Predict unrecorded liabilities using the ML model.
- Combine predictions with actual GR/IR exposure.
- Visualise suppliers to identify financial outliers.

Executive Message

SAP provides supplier spend and open GR/IR balances, while Databricks predicts additional unrecorded liabilities. Comparing spend against total liability helps Finance identify suppliers with disproportionate financial exposure and prioritise investigation before period close.

PO ACCRUAL COMPLETENESS

in %

40.36%

ACCRUED COUNT

270

SAP Business Variable	Description
Eligible PO Count	Purchase Orders eligible for accrual
Accrued PO Count	Purchase Orders with accruals posted
Accrual Completion Status	Status indicating whether the PO has been accrued
Reference Period	Reporting period for the KPI
PO Accrual Completeness %	Percentage of eligible POs accrued

PO Accrual Completeness (SAP Terminology)PO Accrual Completeness measures the percentage of purchase order items with a posted Goods Receipt (GR) that have been appropriately accounted for through the accrual or invoice matching process.CalculationPO Accrual Completeness (%) = (Accrued PO Items ÷ Eligible PO Items) × 100DefinitionsEligible PO ItemsAll PO items for which a Goods Receipt (GR) or Service Entry Sheet (SES) has been posted in SAP.These represent goods or services that have been received by the company and therefore create a potential financial liability.Accrued PO ItemsEligible PO items that have been fully processed through the GR/IR clearing process, with corresponding invoices posted and amounts reconciled.These items are considered complete and no longer require accrual monitoring or follow-up.Open PO ItemsEligible PO items with a posted GR/SES but without a fully matched invoice or completed reconciliation.These remain outstanding in the accrual worklist (Silver layer/table) and represent ongoing financial exposure requiring accrual consideration.Business InterpretationA higher PO Accrual Completeness % indicates stronger control over the Procure-to-Pay (P2P) process, ensuring that liabilities arising from received goods and services are accurately recognized and reflected in the financial statements. Conversely, open PO items highlight potential accrual gaps and unresolved GR/IR differences that require attention.

PO Accrual Completeness
SAP Functional Perspective
Business Purpose

Measure the percentage of eligible Purchase Orders (POs) for which accruals have been successfully posted. This KPI helps Finance monitor the completeness of the accrual process before period close.

Business Formula

PO Accrual Completeness (%) = (Accrued PO Count / Eligible PO Count) × 100

Business Logic

- Identify all eligible Purchase Orders.
- Identify Purchase Orders with accruals posted.
- Count eligible and accrued Purchase Orders.
- Calculate the accrual completeness percentage.
- Display KPI cards and bucket the results (No Invoice, Low, Partial, Near Complete).

CFO Business Value

- Measures the completeness of the accrual process.
- Highlights gaps before financial close.
- Supports proactive follow-up on missing accruals.
- Improves the accuracy and completeness of financial reporting.

PO ACCRUAL COMPLETENESS

in %

40.36%

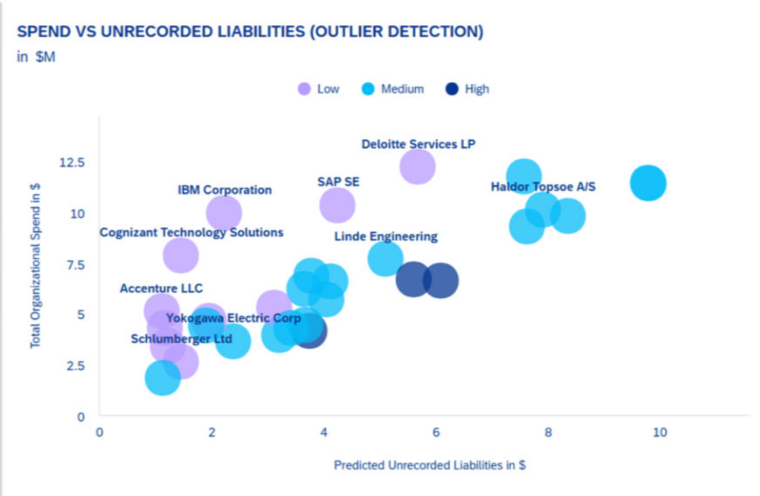
ACCRUED COUNT

270

PO Accrual Completeness
SAP Functional Perspective
Business Purpose
Measure the percentage of eligible Purchase Orders (POs) for which accruals have been successfully posted. This KPI helps Finance monitor the completeness of the accrual process before period close.
Business Formula
PO Accrual Completeness (%) = (Accrued PO Count / Eligible PO Count) × 100
Business Logic
• Identify all eligible Purchase Orders.
• Identify Purchase Orders with accruals posted.
• Count eligible and accrued Purchase Orders.
• Calculate the accrual completeness percentage.
• Display KPI cards and bucket the results (No Invoice, Low, Partial, Near Complete).
CFO Business Value
• Measures the completeness of the accrual process.
• Highlights gaps before financial close.
• Supports proactive follow-up on missing accruals.
• Improves the accuracy and completeness of financial reporting.

PO Accrual Completeness (SAP Terminology)PO Accrual Completeness measures the percentage of purchase order items with a posted Goods Receipt (GR) that have been appropriately accounted for through the accrual or invoice matching process.
Calculation PO Accrual Completeness (%) = (Accrued PO Items ÷ Eligible PO Items) × 100
Definitions Eligible PO Items All PO items for which a Goods Receipt (GR) or Service Entry Sheet (SES) has been posted in SAP. These represent goods or services that have been received by the company and therefore create a potential financial liability.
Accrued PO Items Eligible PO items that have been fully processed through the GR/IR clearing process, with corresponding invoices posted and amounts reconciled. These items are considered complete and no longer require accrual monitoring or follow-up. Open PO Items Eligible PO items with a posted GR/SES but without a fully matched invoice or completed reconciliation. These remain outstanding in the accrual worklist (Silver layer/table) and represent ongoing financial exposure requiring accrual consideration. Business Interpretational higher PO Accrual Completeness % indicates stronger control over the Procure-to-Pay (P2P) process, ensuring that liabilities arising from received goods and services are accurately recognized and reflected in the financial statements. Conversely, open PO items highlight potential accrual gaps and unresolved GR/IR differences that require attention.

APPENDIX



Vendor Exposure vs Invoice Delay — Bubble Chart

Definition: A scatter plot that maps each supplier's total open accrual exposure against their average invoice delay (days since GR without full matching), with bubble size representing the volume of open items — revealing which vendors combine high financial risk with slow invoice turnaround.

Formulas:

Per Supplier: X-axis: Avg_Invoice_Delay = AVG(Days_Open) Y-axis: Total_Exposure = SUM(Exposure_Amount_USD) Bubble: Open_Items = COUNT()

Color: POST_Rate = COUNT(Recommended_Action = 'POST') / COUNT()

Where: Days_Open = Reference_Date − Oldest_GR_Date

Colour Logic:

Condition	Colour	Meaning
POST Rate > 50%	Teal	High Risk — majority of items require immediate accrual posting
POST Rate ≤ 50%	Pink	Normal — items under review or observation

High Risk Zone (red shaded quadrant):X_Threshold = MEDIAN(Avg_Days_Open) across all suppliers Y_Threshold = MEDIAN(Total_Exposure) across all suppliers

High Risk = Supplier in upper-right quadrant (above BOTH medians)Interpretation:

Quadrant	Meaning	Action
Upper-right (red zone)	High exposure + long delays — highest priority	Immediate supplier escalation
Upper-left	High exposure but fast processing — volume issue	Monitor capacity
Lower-right	Low exposure but slow invoicing — process issue	Supplier follow-up on invoicing SLA
Lower-left	Low exposure + fast — healthy	No action

Business use: Answers "Which suppliers combine high financial exposure with slow invoice submission?" — enables prioritisation by simultaneously assessing both the monetary risk and the operational delay per vendor.

Spend vs Unrecorded Liabilities — Scatter (Outlier Detection)

Definition: Plots each supplier by their total organizational spend against their total predicted unrecorded liabilities, with colour intensity representing the exposure index — revealing outlier vendors whose liability-to-spend ratio deviates significantly from the portfolio norm.

Formulas:Per Supplier: X-axis: Total_Organizational_Spend = Total_IR_Amount + Predicted_Unrecorded_Liabilities Y-axis: Predicted_Unrecorded_Liabilities = GR_IR_Accrual + Predicted_No_Doc_Accrual Colour: Exposure_Index_% = (Predicted_Unrecorded_Liabilities / Total_Organizational_Spend) × 100

Where: GR_IR_Accrual = SUM(Exposure_Amount_USD) — known open GR/IR balance Predicted_No_Doc_Accrual = ML-predicted liability for items with no GR posted Total_IR_Amount = SUM(all matched invoices in USD)Colour Gradient:

Colour	Index Range	Meaning
Red (hot)	> 80%	Nearly all spend unresolved — critical
Yellow (warm)	50–80%	Significant unresolved portion — monitor
Green (cool)	< 50%	Healthy — most spend matched

Outlier Detection Logic:

Upper-right quadrant: High spend + high liabilities — critical vendors (large AND problematic)

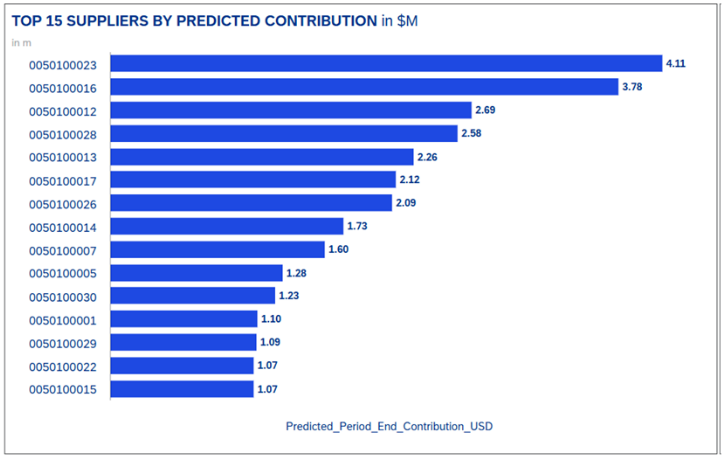
Upper-left quadrant: Low spend + high liabilities — PO/contract anomalies (small vendor, big liability = suspicious)

Lower-right quadrant: High spend + low liabilities — well-managed large vendors

Lower-left quadrant: Low spend + low liabilities — no concern

Diagonal reference (implicit): Suppliers on the 45° line have 100% exposure index (all spend is unrecorded). Points below the diagonal have some healthy invoiced spend.

Business use: Answers "Is risk proportional to vendor size, or are there outliers?" — identifies vendors whose liabilities are disproportionate to their spend volume, signalling potential contract disputes, systematic GR failures, or anomalous procurement patterns.



Predicted Period End Accrual — ML Pipeline
KPI Definition: Single most visible accrual number the Controller and Chief Financial Officer look for at close — the total predicted GR/IR accrual balance remaining at period end.
Formula:
Predicted Period End Accrual = Σ (Exposure_Amount × P(Still Open at Close)) across all active accrual lines
P(Still Open at Close) — Random Forest Classifier output: probability a given accrual line will NOT be resolved (invoiced) before fiscal period end
Confidence Band = Mean ± 1.96 × Std of predicted probabilities (95% interval)
ML Approach: Random Forest binary classifier (sklearn) trained on historical period snapshots. Labels: 1 = accrual line persisted to period end, 0 = resolved before close. Features: log exposure, PO attributes, supplier reliability, aging bucket, fiscal timing. The predicted probabilities weight each open line's exposure to produce the aggregate forecast.
CFO Impact: Replaces manual spreadsheet estimates · enables mid-period forecasting · reduces post-close surprises · provides confidence intervals for audit readiness

Top 15 Suppliers by Predicted Contribution
Definition: Ranks the 15 suppliers with the highest dollar contribution to the predicted period-end accrual — identifying which vendors are driving the largest share of unresolved liabilities that the model expects to persist at fiscal close.
Formula:
Supplier_Contribution = SUM(Predicted_Period_End_Contribution_USD) GROUP BY Supplier ORDER BY Supplier_Contribution DESC LIMIT 15
Where per item: Predicted_Period_End_Contribution_USD = Exposure_Amount_USD × Avg_P_Still_Open
Logic:
For each open PO line item, multiply its exposure by the model's persistence probability
Group by Supplier and sum the weighted contributions
Sort descending by total predicted contribution
Take top 15
Business use: Turns the headline accrual number into an actionable supplier-by-supplier hit list. Answers: "Which vendors should I call first to reduce the accrual before close?"

PREDICTED PERIOD END ACCRUAL in $

38.50 m

Predicted Period End Accrual Definition :

Predicted Period End Accrual = SUM (Predicted contribution of each item)

Predicted Period End Contribution USD = Exposure Amount USD × Avg P Still Open

Meaning: Each open accrual line's exposure is weighted by the model's predicted probability that it will persist (not be invoiced) by period close. Summing across all lines gives the total forecasted accrual balance.

Model Features Used Feature Definition

REFERENCE_DATE – Oldest GR Date (aging)

Supplier Reliability

Historical on-time invoice rate (0–1)

Invoice Coverage

Signed IR Amount / Signed GR Amount (0–2)

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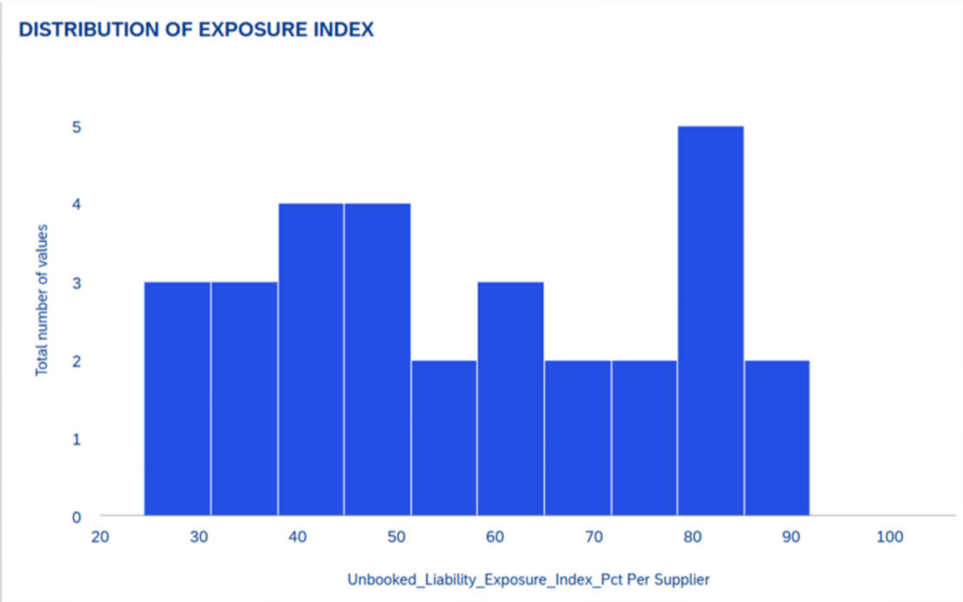
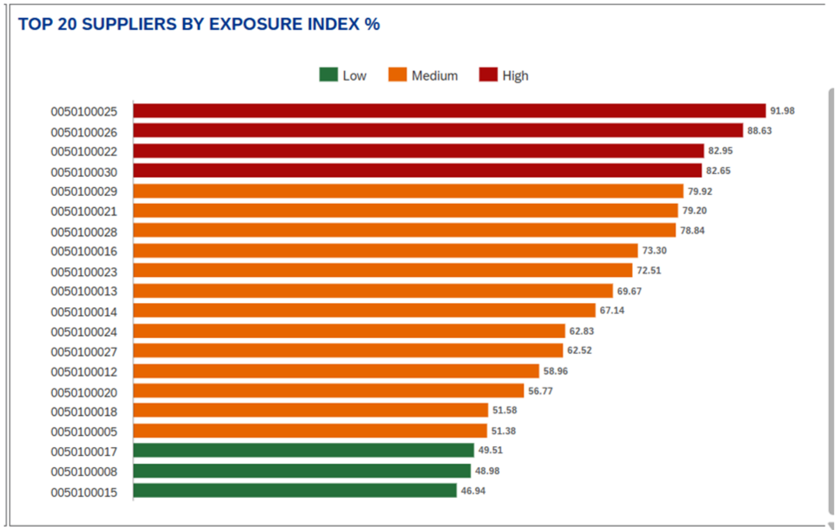


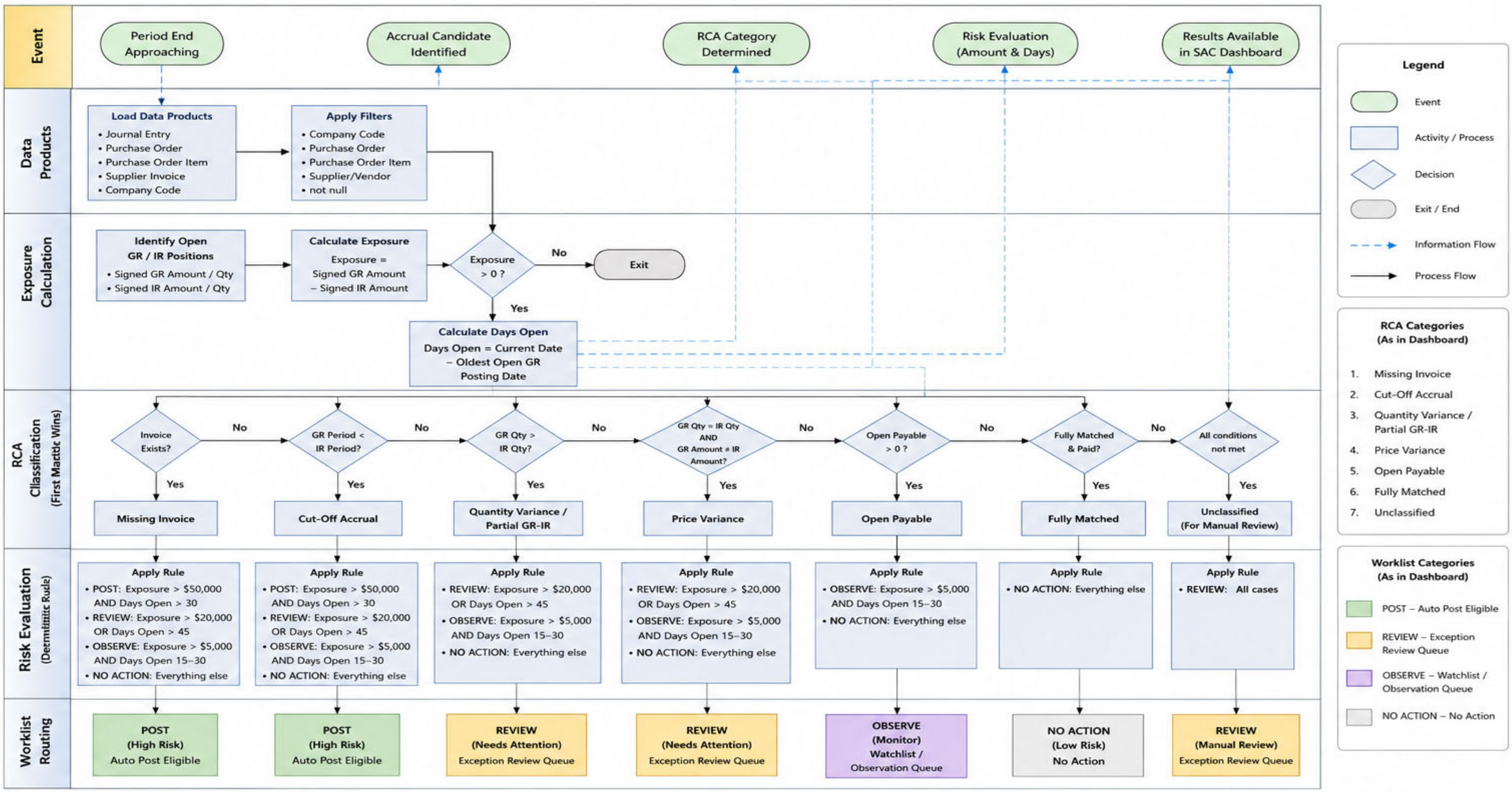
Unbooked Liability Exposure Index — ML Pipeline
KPI Definition: Estimated liabilities likely incurred but not yet recorded before invoice receipt.
Formula:
Unbooked Liability Exposure Index (%) = (Predicted Unrecorded Liabilities ÷ Total Organizational Spend) × 100
Predicted Unrecorded Liabilities = GR/IR Accrual + Predicted No-Document Accrual
GR/IR Accrual = MAX(Signed_GR_Amount − Signed_IR_Amount, 0) — rule-based, from silver fact_accrual_worklist
Predicted No-Document Accrual — ML model predicting obligations for POs with **no GR posted**
Total Organizational Spend = Σ(Signed_IR_Amount across all invoices) + Predicted Unrecorded Liabilities
ML Approach: LightGBM Regressor trained on historical GR/IR accruals using only PO-level features available *before* goods receipt (no leakage). Applied to open POs with no GR to estimate hidden liabilities.
CFO Impact: Early warning of hidden balance sheet risk · improves liability forecasting · reduces post-close surprises · increases financial statement completeness

Distribution of Exposure Index (Histogram) — Theoretical DefinitionDefinition: A frequency distribution showing how many suppliers fall into each exposure index percentage band, revealing whether unbooked liability risk is concentrated in a few vendors or spread systemically across the portfolio.Formula:Per Supplier: Exposure_Index = (Predicted_Unrecorded_Liabilities / Total_Organizational_Spend) × 100Where: Predicted_Unrecorded_Liabilities = GR_IR_Accrual + Predicted_No_Doc_Accrual Total_Organizational_Spend = Total_IR_Amount + Predicted_Unrecorded_LiabilitiesHistogram: Bin_Width = 5% Bin_Assignment = FLOOR(Exposure_Index / 5) × 5 Frequency = COUNTD(Supplier) per bin Median = MEDIAN(Exposure_Index) across all suppliersThresholds:x = 80% — High Risk (red line)x = 50% — Medium Risk (orange line)x = Median — Portfolio centre (teal line)Interpretation:Shape MeaningLeft-skewed (peak at 0-30%) Healthy — most suppliers well-managed, few outliersRight-skewed (peak at 70-100%) Systemic failure — widespread GR/IR discipline breakdownFlat / uniform No clear pattern — inconsistent controls across portfolioBimodal (two peaks) Two distinct populations — requires segmented strategiesBusiness use: Answers "Is this a few bad apples or a rotten barrel?" Track the distribution shape quarter-over-quarter to measure whether process improvements are shifting suppliers leftward (toward lower risk).



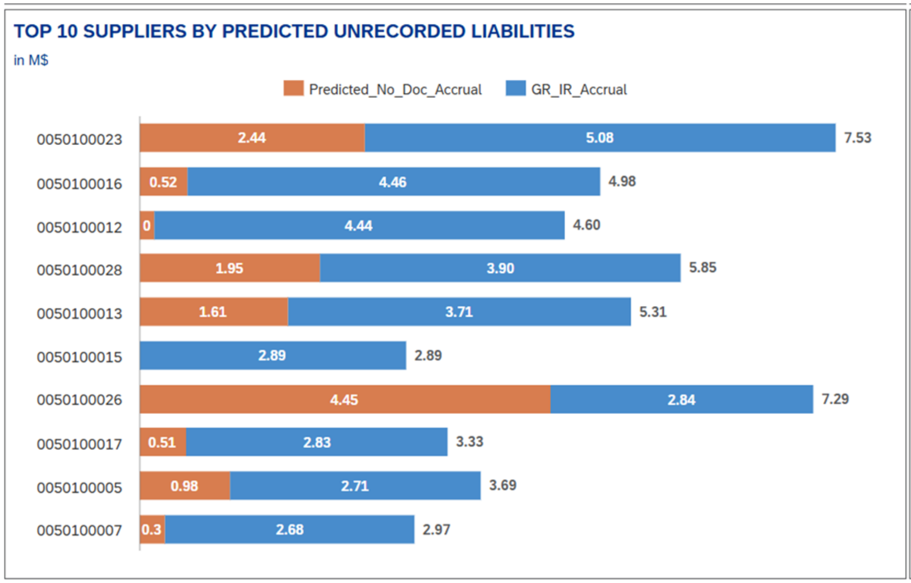
Top 20 Suppliers — Exposure Index % (Conditional Bar)Definition: Ranks the 20 suppliers with the highest Unbooked Liability Exposure Index — showing what percentage of each supplier's total financial relationship is tied up in unresolved or unrecorded liabilities, colour-coded by risk severity.Formula:Per Supplier: Exposure_Index_% = (Predicted_Unrecorded_Liabilities / Total_Organizational_Spend) × 100Where: Predicted_Unrecorded_Liabilities = GR_IR_Accrual + Predicted_No_Doc_Accrual Total_Organizational_Spend = Total_IR_Amount + Predicted_Unrecorded_LiabilitiesRanking: ORDER BY Exposure_Index_% DESC LIMIT 20Colour Logic: Index Range Colour Risk Level Meaning> 80% Crimson High Nearly all spend is in unresolved liabilities — possible contract disputes, systematic GR failures, or fraud risk50–80% Dark Orange Medium Significant portion uninvoiced — needs monitoring and follow-up< 50% Steel Blue Low Healthy invoice coverage — majority of spend properly matched Business use: Answers "For which suppliers is most of our financial relationship sitting in unresolved liabilities?" A supplier at 90% means nearly all spend is unrecorded — possible indicators include contract disputes, systematic GR failures, or fraud risk. Provides instant prioritisation for the CFO's action list.

AI Driven Accrual Worklist Process Flow



Notes:

- First match wins in RCA Classification.
- Risk Evaluation uses deterministic rules based on Exposure Amount and Days Open.
- Worklist categories drive dashboard buckets in SAC.



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GR/R Accrual = MAX(Signed_GR_Amount − Signed_IR_Amount, 0) — rule-based, from silver fact_accrual_worklist
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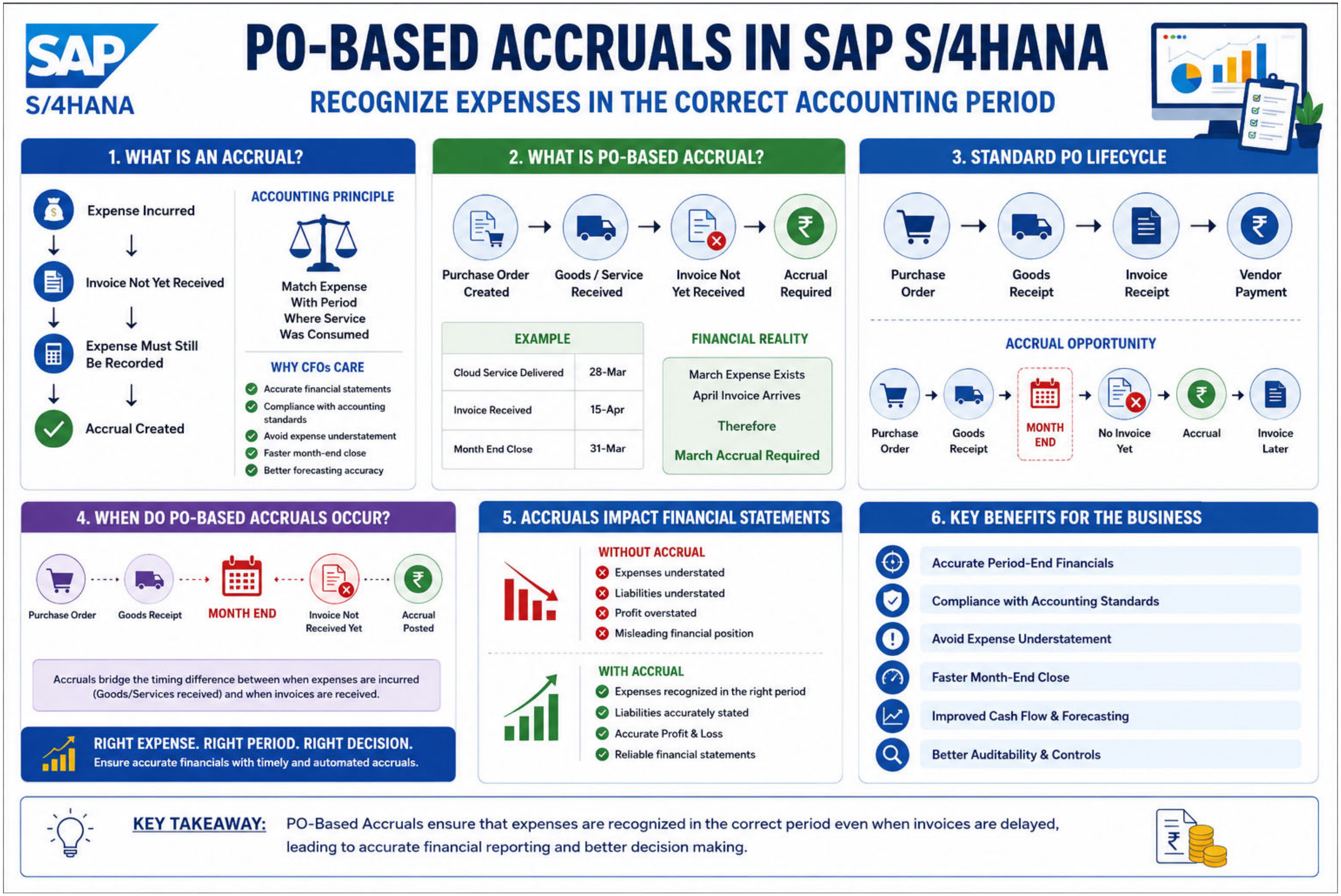
CFO Impact: Early warning of hidden balance sheet risk · improves liability forecasting · reduces post-close surprises · increases financial statement completeness

Top 10 Suppliers — Predicted Unrecorded Liabilities (Stacked Bar)Definition: Ranks the 10 suppliers with the highest total predicted unrecorded liabilities, split into the known rule-based GR/IR accrual and the ML-predicted no-document accrual — showing both visible and hidden liability components per vendor.Formula:Per Supplier: GR_IR_Accrual = SUM(Exposure_Amount_USD) for items WITH Goods Receipt posted Predicted_No_Doc_Accrual = SUM(ML-predicted exposure) for items with NO Goods Receipt Total_Unrecorded = GR_IR_Accrual + Predicted_No_Doc_AccrualRanking: ORDER BY Total_Unrecorded DESC LIMIT 10Where:GR_IR_Accrual — the known, rule-based accrual already on the balance sheet (GR posted, invoice not yet matched)Predicted_No_Doc_Accrual — ML-predicted liability for POs where no GR exists yet (hidden risk invisible to standard AP ageing)Stacked components:Component Meaning ColourGR/IR Accrual (rule-based) Known liability — GR posted, invoice pending BluePredicted No-Doc Accrual (ML) Hidden liability — no GR posted, model predicts exposure will materialise OrangeInterpretation:Pattern Meaning ActionSupplier dominated by blue (GR/IR) Traditional accrual issue — invoice delays AP follow-up for invoice submissionSupplier dominated by orange (No-Doc) Missing Goods Receipts — hidden liability Procurement follow-up to locate/post GRsLarge total bar vs. peersConcentrated risk in single vendor CFO/CPO escalationBusiness use: Answers "Which suppliers carry the largest total financial exposure — and is that risk visible or hidden?" Suppliers dominated by the orange (ML-predicted) bar need urgent procurement attention to locate missing goods receipts before period close.



SAP
S/4HANA
PO-BASED ACCRUALS IN SAP S/4HANA
RECOGNIZE EXPENSES IN THE CORRECT ACCOUNTING PERIOD
1. WHAT IS AN ACCRUAL?
Expense Incurred
Invoice Not Yet Received
Expense Must Still Be Recorded
Accrual Created
ACCOUNTING PRINCIPLE
Match Expense With Period Where Service Was Consumed
WHY CFOs CARE
Accurate financial statements
Compliance with accounting standards
Avoid expense understatement
Faster month-end close
Better forecasting accuracy
2. WHAT IS PO-BASED ACCRUAL?
Purchase Order Created
Goods / Service Received
Invoice Not Yet Received
Accrual Required
EXAMPLE
Cloud Service Delivered 28-Mar
Invoice Received 15-Apr
Month End Close 31-Mar
FINANCIAL REALITY
March Expense Exists
April Invoice Arrives
Therefore
March Accrual Required
3. STANDARD PO LIFECYCLE
Purchase Order
Goods Receipt
Invoice Receipt
Vendor Payment
ACCRUAL OPPORTUNITY
Purchase Order
Goods Receipt
MONTH END
No Invoice Yet
Accrual
Invoice Later
4. WHEN DO PO-BASED ACCRUALS OCCUR?
Purchase Order
Goods Receipt
MONTH END
Invoice Not Received Yet
Accrual Posted
Accruals bridge the timing difference between when expenses are incurred (Goods/Services received) and when invoices are received.
RIGHT EXPENSE. RIGHT PERIOD. RIGHT DECISION.
Ensure accurate financials with timely and automated accruals.
5. ACCRUALS IMPACT FINANCIAL STATEMENTS
WITHOUT ACCRUAL
Expenses understated
Liabilities understated
Profit overstated
Misleading financial position
WITH ACCRUAL
Expenses recognized in the right period
Liabilities accurately stated
Accurate Profit & Loss
Reliable financial statements
6. KEY BENEFITS FOR THE BUSINESS
Accurate Period-End Financials
Compliance with Accounting Standards
Avoid Expense Understatement
Faster Month-End Close
Improved Cash Flow & Forecasting
Better Auditability & Controls
KEY TAKEAWAY: PO-Based Accruals ensure that expenses are recognized in the correct period even when invoices are delayed, leading to accurate financial reporting and better decision making.

SAP
S/4HANA

PO ACCRUALS IN SAP FI – POSTING LIFECYCLE

RECOGNIZE NOW, SETTLE LATER

MONTH-END ACCRUAL POSTING LIFECYCLE

1. PERIOD END

Goods Received
Invoice Missing

ACCRUAL POSTED

Account	Debit (₹)	Credit (₹)
Expense Account	100,000	
Accrued Liability (Current Liability)		100,000

Accrual is recorded to recognize expense in the current period.

2. BEGINNING OF NEXT MONTH

Automatic
Reversal

ACCRUAL REVERSAL

Account	Debit (₹)	Credit (₹)
Accrued Liability (Current Liability)	100,000	
Expense Account		100,000

Reversal removes the accrual entry from the new period.

3. INVOICE ARRIVES

Invoice
Posting

INVOICE POSTING

Account	Debit (₹)	Credit (₹)
Expense Account	100,000	
Vendor Liability (Accounts Payable)		100,000

Actual invoice is posted and liability to vendor is recorded.

4. VENDOR PAYMENT

Payment
to Vendor

PAYMENT POSTING

Account	Debit (₹)	Credit (₹)
Vendor Liability (Accounts Payable)	100,000	
Bank Account		100,000

Vendor liability is cleared when payment is made.

KEY TAKEAWAYS

- Accrual is posted at period end when invoice is missing.
- System automatically reverses the accrual in the next period.
- Invoice posting replaces the accrual entry.
- Payment clears the vendor liability.
- Ensures accurate expense recognition in the right period.

BUSINESS IMPACT

- Accurate Financial Statements (Right Expense in Right Period)
- Compliance with Accounting Standards
- Avoid Expense Understatement
- Better Cash Flow & Forecasting
- Faster Month-End Close
- Audit Ready & Transparent

SAP FI VIEW – END TO END FLOW

Period End (Accrual Posting) → Automatic Reversal → Invoice Posting → Vendor Payment → Financial Close Completed

KEY ACCOUNTS USED

Expense Account	Recognizes the expense in the period of goods / services received
Accrued Liability (Current Liability)	Temporary liability created for accrued expenses
Vendor Liability	Liability recorded when invoice

WHAT HAPPENS AFTER REVERSAL?

After automatic reversal in the next period:

No accrual balance → System is ready → Invoice posting → Books reflect true

COMMON SAP DOCUMENT TYPES

Accrual Posting (Manual / Automatic)	Journal Entry
Accrual Reversal (Automatic)	Reversal Entry

SAP S/4HANA

AI-DRIVEN PO ACCRUAL USE CASE

FIND. CLASSIFY. PRIORITIZE. RECOMMEND.

BUSINESS CHALLENGE

Among thousands of open POs, which ones need Accruals, which need review, which need monitoring, and which need no action?

- Large volume of transactions
- Manual review is time consuming
- Higher risk of missed accruals
- Inconsistent judgments
- Audit & compliance risk
- Slower month-end close

OUR SOLUTION APPROACH

Connect SAP & External Data in Databricks → Build Data Products → Calculate Exposure & Days Open → Classify Root Cause (RCA) → AI Prioritization & Recommendation → Actionable Insights in SAC Dashboard

DATA PRODUCTS IN DATABRICKS

- Purchase Order (Header, Line, Terms)
- Journal Entry (ACDOCA)
- Goods Receipt (MSEG)
- Supplier Invoice (RBKP, RSEG)
- Company / Vendor / Material / Calendar

RCA CLASSIFICATION – IDENTIFY THE TRUE SCENARIO

	CONDITION	OUTCOME
1 MISSING INVOICE	Goods Received (GR) exists No Invoice (IR) exists	Missing Invoice
2 CUT-OFF ACCRUAL	GR Posting Date < IR Posting Date	Cut-Off Accrual
3 QUANTITY VARIANCE (PARTIAL GR-IR)	GR Quantity > IR Quantity	Quantity Variance
4 PRICE VARIANCE	GR Quantity = IR Quantity AND GR Amount ≠ IR Amount	Price Variance
5 OPEN PAYABLE	Invoice exists Invoice not paid Open Payable > 0	Open Payable
6 FULLY MATCHED	GR = IR Invoice Paid Open Payable = 0	Fully Matched
7 UNCLASSIFIED	All other cases not meeting above conditions	Unclassified (Manual Review)

KEY KPI'S DELIVERED

- Total Accrual Exposure (₹)
- Open Accrual Count
- Average Days Open
- Exposure > 30 / 60 / 90 Days
- Auto Post Eligible %
- Review Queue %
- Watchlist %
- No Action %
- Vendor Delay Risk
- Accrual Accuracy Improvement
- Manual Effort Saved (Hours)

AI PRIORITIZATION – RISK SCORING MODEL

INPUT FACTORS

- Exposure Amount (₹)
- Days Open
- Vendor Behavior (Historical Delays)
- Historical Invoice Delay Trend
- Quantity Variance Trend
- Price Variance Trend
- Purchase Order Aging
- Material / Service Criticality

→ **AI MODEL** Risk Score (0 – 100) →

RISK SCORE BAND	
> 85	High
60 – 85	Medium High
30 – 60	Medium
< 30	Low

WORKLIST CATEGORIES (AS IN SAC DASHBOARD)

	AUTO POST ELIGIBLE	EXCEPTION REVIEW QUEUE	WATCHLIST / OBSERVATION QUEUE	NO ACTION
	High confidence accrual candidates	Requires finance review & validation	Monitor only	Low risk / No accrual required
WHEN	• Exposure > ₹50,000 AND • Days Open > 30	• Exposure > ₹20,000 OR • Days Open > 45	• Exposure > ₹5,000 AND • Days Open 15 – 30	• Everything else (Low Risk)
ACTION	POST	REVIEW	OBSERVE	NO ACTION

TECHNOLOGY STACK

SAP S/4HANA · databricks · Delta Lake · AI/ML Model · SAP Analytics Cloud

INTELLIGENT ACCRUALS. RIGHT PERIOD. RIGHT DECISION.

AI-DRIVEN WORKLISTS HELP FINANCE FOCUS ON WHAT MATTERS MOST!